

Precision Agriculture with Remote Sensing and Machine Learning for Indian Agriculture

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Project Domain

Precision Agriculture / Remote Sensing / Machine Learning

Abstract & Problem Statement

Vegetation indices like NDVI are widely used for crop monitoring and yield estimation[1]. For example, Tamayo-Vera et al. (2025) identified NDVI and climate variables as top predictors in a regional potato yield model[2], underscoring the value of satellite-derived vegetation data. In India—where staple crops such as rice (Kharif season) and wheat (Rabi season) dominate—accurate yield forecasting is crucial for food security and policy planning. Satellite platforms (Sentinel-2, Landsat, MODIS) provide open multispectral imagery over India, and national sensors (e.g. Resourcesat-2 AWiFS) also deliver NDVI products. Table: Comparison of remote sensing data sources (below) shows the spatial/temporal resolutions and access characteristics of key datasets applicable to Indian agriculture. These data can be processed in Python (e.g. via Google Earth Engine or GDAL) and fed into deep learning frameworks like TensorFlow for modeling.

Dataset / Source	Bands / Data	Spatial Res.	Temporal Res.	Coverage	Access
Sentinel-2 MSI (Copernicus)	Multispectral (VNIR, SWIR)	10–60 m	5-day revisit (2-satellites)	Global (incl. India)	Free (ESA)
Landsat 8/9 (USGS)	Multispectral (VNIR, Pan)	15 m (Pan), 30 m	16-day (with Landsat 9, ~8-day effective)	Global	Free (USGS)
MODIS (Terra/Aqua, NASA)	Vegetation indices (NDVI/EVI 250 m);	250–1000 m	Daily (composited 8/16-day)	Global (daily)	Free (NASA)

Dataset / Source	Bands / Data	Spatial Res.	Temporal Res.	Coverage	Access
	Reflectance (500 m–1 km)				
Resourcesat-2 AWiFS (ISRO)	Multispectral (Green, Red, NIR)	56 m	~5 days	India & neighbors	Free (Bhuvan)
PlanetScope (Planet Labs)	Multispectral RGB + NIR	3–5 m	~Daily (sun-synchronous)	Global (commercial)	Proprietary (API)

Spatio-temporal machine learning models can capture seasonal dynamics in these data. However, in India frequent cloud cover during the monsoon leads to missing data in optical imagery. Recent research (e.g. Zhang et al. 2024) shows that fusing high-temporal MODIS with higher-resolution Landsat can effectively fill such gaps[3]. We will extend this idea by implementing a **spatio-temporal GAN** (generative adversarial network) to synthesize realistic NDVI time-series for Indian croplands. The generative model (following Li et al., 2023) aims to generate missing NDVI frames and improve image quality. Figure below illustrates this concept: the synthetic map (right) covers the full region whereas the original satellite map (left) has many gaps[3]. These generated NDVI sequences, combined with weather and soil data, will feed into a TensorFlow-based LSTM/Convolutional model to predict crop yields. Python and TensorFlow will be used throughout for data processing, model development, and training.

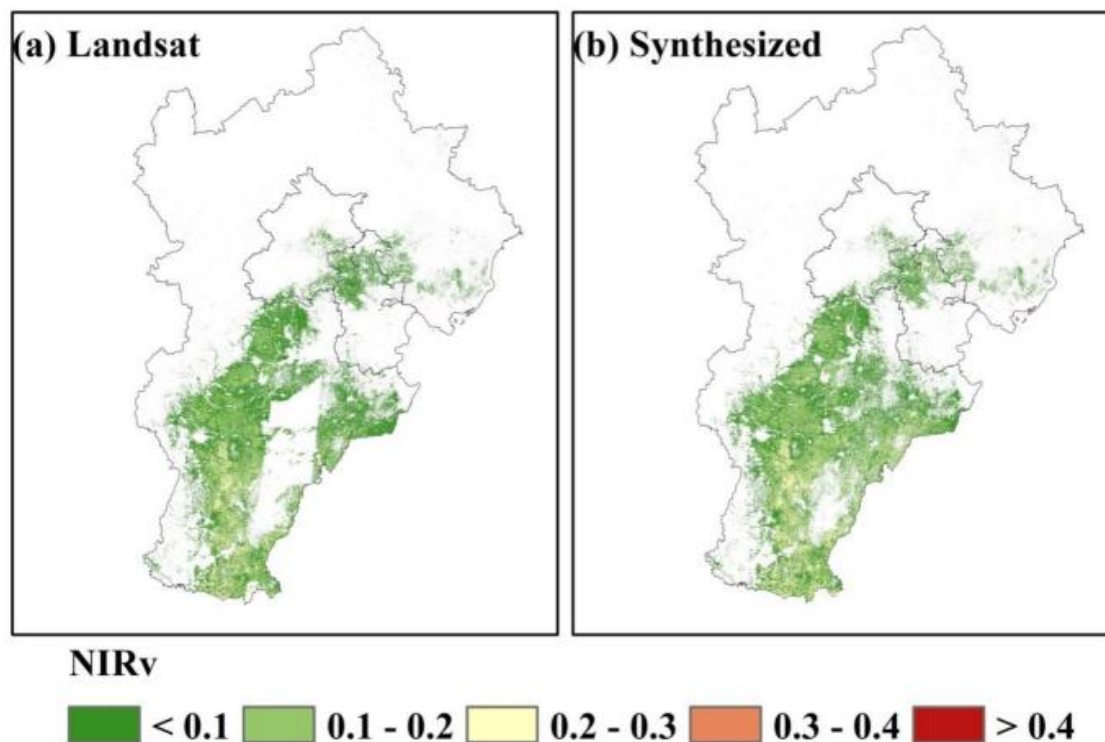


Figure: Example of synthetic vegetation index mapping. (a) Original Landsat-derived index

with missing areas (white) due to clouds; (b) Synthesized map (from fused MODIS+Landsat) with filled-in coverage[3].

Research Gaps Identified

- **India-Specific Modeling:** Most published crop-yield ML models focus on regions like China or the U.S.; there is a gap in tailored models for diverse Indian agro-climatic conditions.
- **Data Scarcity & Gaps:** Frequent cloud cover in India causes missing NDVI data; existing Indian yield-prediction studies seldom use generative methods to fill these gaps.
- **Advanced Index Usage:** Studies often rely on NDVI alone; there is limited evaluation of newer indices (e.g. NIRv, kNDVI) or GAN-augmented imagery for Indian crops.
- **Multi-Source Integration:** Few works integrate multi-resolution/sensor data (e.g. merging MODIS, Sentinel, AWiFS) with ML models for yields in India.
- **Limited Ground Truth:** Open, high-resolution yield datasets (e.g. at district or block level) are limited in India, hampering supervised learning.
- **Modeling Tools:** Adoption of TensorFlow/Python-based end-to-end pipelines for precision ag in Indian universities/industry is still emerging.

Objectives of the Project

1. **Data Collection:** Aggregate open-source remote sensing data (Sentinel-2, Landsat, MODIS, AWiFS NDVI) and relevant climate/soil datasets for chosen Indian regions (e.g. Punjab for wheat, West Bengal for rice).
2. **Generative Modeling:** Implement a spatio-temporal GAN (in TensorFlow) to generate synthetic NDVI image sequences, reducing gaps and noise in satellite data.
3. **Yield Prediction Model:** Develop a deep learning model (e.g. CNN-LSTM) that takes the generated NDVI series and auxiliary data (rainfall, temperature) to predict crop yield.
4. **Comparison & Validation:** Compare GAN-augmented model against baselines (e.g. LSTM without GAN, random forest). Evaluate performance on Indian crops and seasons using metrics (R^2 , RMSE).
5. **Toolchain:** Build the pipeline in Python with TensorFlow/Keras, demonstrating end-to-end feasibility for Indian datasets.
6. **Documentation:** Analyze results to identify strengths/limitations of generative augmentation in the Indian context and recommend future improvements.

Unique Contribution

- **India Focus:** This project uniquely applies advanced GAN-based methods to India's agriculture, addressing local crop calendars and monsoon patterns.

- **Generative Data Augmentation:** First use of spatio-temporal GANs to synthesize high-resolution NDVI for Indian yield modeling, potentially improving predictions under cloud cover.
- **Multi-Dataset Fusion:** Combines diverse data sources (Sentinel, MODIS, ISRO sensors) to exploit rich, high-resolution imagery.
- **Modern ML Pipeline:** End-to-end Python/TensorFlow implementation provides a replicable framework for Indian agriculture applications.
- **Policy Relevance:** The project's predictions can aid policymakers by providing early yield forecasts for key crops, leveraging Indian data.

In-house / Out-house

In-house (Departmental project; to be implemented within the university's research group or academic lab).

Sources: Recent studies on crop yield modeling[1][2][3] and available Indian satellite datasets were used to shape this proposal. Data sources and model options are compared in the table above.

[1] [3] Predicting wheat yield from 2001 to 2020 in Hebei Province at county and pixel levels based on synthesized time series images of Landsat and MODIS | Scientific Reports

https://www.nature.com/articles/s41598-024-67109-3?error=cookies_not_supported&code=f7c98be0-9022-455e-84f9-854ed3abc90d

[2] Advanced machine learning for regional potato yield prediction: analysis of essential drivers | npj Sustainable Agriculture

https://www.nature.com/articles/s44264-025-00052-6?error=cookies_not_supported&code=e72cdc01-92b2-41b4-b932-1dfc4cac8580